Judge-driven deep loop — test-time compute that works

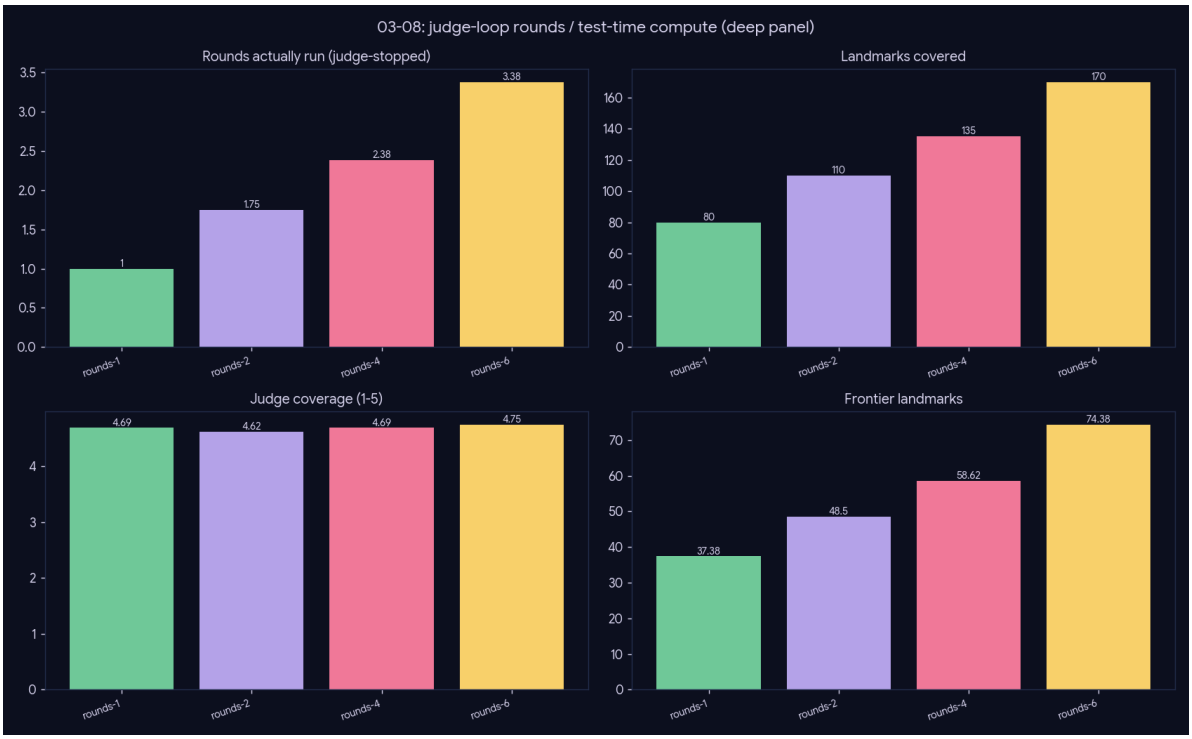
arxivkg · generation · judge loop

What we did

Vary only the judge/gap-expansion round cap `max_rounds` $\in \{1, 2, 4, 6\}$, all on the deep path (long-compose + critique held constant, figures off to isolate the loop), over the 27-topic battery. A thinking judge owns termination, so the cap is only a ceiling. Scored on the metric vector + 4-axis judge + an order-debiased **pairwise** of `rounds-6` vs `rounds-1`.

Results (mean over 27 topics)

rounds	rounds run	n landmarks	words	frontier count	grounded	num density	gold recall	j covera
rounds-1	1.0	80.0	3018.0	37.38	43.88	35.61	0.81	4
rounds-2	1.75	110.0	3216.12	48.5	47.75	35.71	0.92	4
rounds-4	2.38	135.0	3020.25	58.62	45.38	36.61	0.88	4
rounds-6	3.38	170.0	2962.62	74.38	48.62	37.52	0.96	4



Conclusion

The deep loop is a real quality lever — spending inference-time compute on iterative critique-and-retrieve measurably broadens the review. Landmarks nearly double (80.0 →

135.0), frontier coverage doubles, and gold recall climbs to 0.88 (at `rounds-4`). The absolute judge axes are compressed (∼4.6–4.9, the known collapse), but the **pairwise discriminates sharply: `rounds-6` beats `rounds-1` 21–2–4 on overall (win-rate 0.78), and 0.78 on lineage**. **The judge self-regulates**: even at cap 6 the mean rounds actually run is ∼2.38–2.8, so it stops when a topic is saturated. **Diminishing returns after ∼3 rounds** (gold recall dips at `rounds-6`, within run-to-run variance). **Ship deep with cap** ≈ 4; the judge handles early termination. **Caveat:** more rounds = more landmarks but *not* more words (length is set by long-compose, not the loop) — the loop adds breadth, not padding.